SOFTWARE REQUIREMENTS

Chat & Messaging Module

Community Hub App

Academic Year 2025–2026

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1. Project Overview

The Community Hub is a collaborative application built by four student teams. Its goal is to connect people through a community feed, collaborative events, a chatting section and a matchmaking one.
Our team (TSO) owns the **Chat & Messaging** module, covering all real-time communication: direct messages, group conversations, and the social layer connecting users.
This document describes the **complete functional requirements** for this module. It covers all planned features and a section of **potential** future enhancements under consideration.

2. User Accounts

2.1 Home Page – Owner: Rares

- When someone opens the module while not logged in, they land on a home page that presents two options: Log In and Sign Up.
- Neither the conversations nor any other feature can be accessed until the user has logged in.

2.2 Log In – Owner: Rares

- On the Log In page, the user enters their username and password to get into their account.
- A "Forgot password?" option is available for anyone who needs to recover access to their account.
- Once logged in, the user can now access the module's features.
- Passwords are always hidden when typed, anywhere in the app, each character being represented by an asterisk. (*)
- If the user last chose their status to be "Offline" ("Invisible") or "Busy", it remains this way when logging in. Otherwise, the user's status is set to "Online".

2.3 Sign Up – Owner: Rares

- To create a new account, the user fills in the following details: Username, Password, Email address, Phone number, and Birthday.
- The username must be at least 5 characters long and at most 16. Each username must be unique.
- The password must be at least 8 characters long and at most 32. The password must include at least one uppercase letter, one number and one special character in order to be considered valid.
- The Email address must be of a valid format, containing exactly one "@" character, a non-empty local part, a valid domain containing at least one dot, and no whitespace characters.

- The phone number must contain only digits and an optional leading '+', having between 7 and 15 characters.
- When entering their date of birth, the user shall select a past date using a calendar picker.
- Once signed up, the user gets logged in automatically and their initial status is "Online".

2.4 Forgot Password – Owner: Rares

- If someone forgets their password, they may choose to reset it by using their email address.
- This sends a one-time code to their email, which they are required to input afterwards to access a password reset form.
- Once they confirm a valid password, their account password is updated with it.

2.5 Log Out – Owner: Rares

- The user may log out of their account at any time, automatically setting their status to "Offline."
- This shall be done by a "Log out" option available on the user's profile.

3. User Profiles

3.1 Viewing Your Own Profile – Owner: Eduard

- Users can view their own profile, which shows their username, profile picture, bio, current status, and birthday.
- The user's profile picture must be of JPEG/PNG format, with a file size smaller than 5MB, and a square aspect ratio.
- The user may add/modify their bio at any time in their profile. It may contain up to 100 characters.
- Users can update their bio, profile picture, and birthday at any time.
- Users can switch their status between Online, Offline, and Busy whenever they like.

3.2 Viewing Other Users' Profiles – Owner: Eduard

- Users can open anyone else's profile to see their username, profile picture, bio, current status, and birthday.
- The profile shows how many mutual friends the two users share.
- From the profile, users can send a friend request or block that person.

3.3 Online Presence – Owner: Eduard

- Each user's status (Online, Offline, or Busy) is visible everywhere in the app: in the friends list, user list, group member panels, and their profiles.
- Status changes appear across the app in near real time.
- Users can choose to appear Offline even while actively using the app, effectively going "invisible".

3.4 Birthday – Owner: Eduard

- On a user's birthday, a small birthday indicator appears next to their name in the friends list, user list, group member panels, and their profiles.

3.5 Mutual Friends – Owner: Eduard

- Interacting with the number of mutual friends shows the full list of shared friends.

4. Friends & Social Connections

4.1 Sending & Managing Friend Requests – Owner: Fineas

- Users can send a friend request to anyone from the user list.
- A dedicated tab shows all incoming friend requests, each with Accept and Decline options.
- Accepting a request automatically opens a direct message conversation between the two users, with a colored border that disappears once the conversation is opened for the first time.
- Declining a request removes it.

4.2 Friends List – Owner: Fineas

- Users can see all their current friends, with each entry showing their profile picture, username, and current status.
- Users can open a direct message with any friend directly from this list.

4.3 Blocking – Owner: Fineas

- Users can block someone from their profile, a direct message conversation, or the group member panel.
- Blocking prevents the blocked user from sending direct messages to the person who blocked them.
- Blocked users can still appear in shared groups but cannot interact privately with the person who blocked them.

5. Conversation List

5.1 Tabs – Owner: Alex

- The conversation list includes the following tabs: All, Direct Messages, Groups, Favorites, and Unread.
- The Unread tab shows every conversation with at least one unread message.
- The Favorites tab shows all conversations marked as favorites.

5.2 Conversation Preview – Owner: Alex

- Each entry shows the contact's name or avatar, a preview of the most recent message, the time it was sent, and an unread message count badge.
- Conversations with unread messages are visually highlighted.
- If the last message was deleted, the preview shows a message accordingly.

5.3 Favorites – Owner: Alex

- Users can mark any conversation as a favorite.
- They may also unmark it from favorites at any time.

5.4 Search – Owner: Alex

- A search bar lets users filter conversations by Group name or by someone’s username (for DMs).

5.5 Unread Separator: Alex

- When the user opens a conversation with unread messages, it scrolls automatically to the first unread message by the user.
- A line separates the messages that had been seen already and the ones that are unread, also specifying the number of unread messages.

6. Direct Messages (DMs)

6.1 DMs Condition – Owner: Andrei

- A user cannot send messages to someone who is not in their friends list.

6.2 DM Conversation View – Owner: Andrei

- Messages appear in chronological order, each showing the sender's profile picture, content, and timestamp.
- Edited messages display an "edited" label next to the timestamp.
- Deleted messages show "This message was deleted" in place of the original content.

6.3 Pinned Messages in DMs – Owner: Andrei

- Either user can pin any message in the DM. The pinned message then appears in a banner at the top of the conversation.
- When pinning a message, the user can choose between one week, two weeks, one month or a custom time after which the pin expires.
- Choosing the custom time option lets the user select a date from a calendar and time picker.
- A notice specifying that the message has been pinned appears for both users.
- Only one message can be pinned at a time. Pinning a message while another one is pinned replaces it.
- Either user can unpin a message at any time.

6.4 Blocked DMs – Owner: Andrei

- If the user has been blocked by someone, the message input is disabled for both of them.

7. Group Conversations

7.1 Creating a Group – Owner: Cristi

- Users can create a group by giving it a name, optionally uploading a group icon (otherwise it's going to be a default icon), and selecting at least two other members.
- The person who creates the group automatically becomes the first admin.
- A notice specifying that the group has been created appears for everyone in it.

7.2 Member Panel (Right Side) – Owner: Cristi

- A collapsible panel on the right side of the screen lists all group members with their profile picture, username, current status, and role.
- Admins are clearly labelled within the panel.
- Admins can see when timeout expire for users.
- Admins can see what members are banned from the group.

7.3 Pinned Message – Owner: Cristi

- An admin can pin any message in the group. The pinned message then appears in a banner at the top of the conversation.
- When pinning a message, the admin can choose between one week, two weeks, one month or a custom time after which the pin expires.
- Choosing the custom time option lets the admin select a date from a calendar and time picker.
- A notice specifying that the message has been pinned appears for everyone in the group.
- Only one message can be pinned at a time. Pinning a message while another one is pinned replaces it.
- An admin can unpin a message at any time.

7.4 Custom Nicknames – Owner: Cristi

- Users can set a custom display name for themselves within a specific group.
- This nickname is visible to all members of that group instead of the user's regular username.
- Users could clear their nickname at any time to revert back to their username.
- The nickname's length must be of maximum 16 characters.

7.5 Role Management – Owner: Cristi

- Groups have three roles: admin, member, and banned.
- Admins can promote members to admin, demote admins back to member, apply timeouts, and ban members.
- A notice specifying that a role has been changed appears for everyone in the group.

7.6 Leaving Groups – Owner: Cristi

- Any member can leave a group at any time.
- If the last admin leaves the group, a pop-up appears to promote another member before leaving. This gives them the option to choose from the list of current members.
- If the group becomes completely empty, it is deleted.

8. Messaging

8.1 Sending Messages – Owner: Andrei

- Users can type messages in chats at any time and send them.
- A message may include text and any Unicode Emojis.
- The maximum length of a message is 1024 characters.
- Empty messages cannot be sent.

8.2 Message Editing – Owner: Andrei

- The sender of a message can edit its text content after sending.
- The original text is no longer available once the message has been edited, only the updated version is visible.

8.3 Message Deletion – Owner: Andrei

- The sender of a message (or a group admin) can delete a message.

8.4 Message History & Scrollback – Owner: Andrei

- The full message history is available to all current participants.
- Users who join a group after it was created can still see all messages that were sent before they were members.
- Users can scroll up to load older messages.

8.5 Image Attachments – Owner: Andrei

- Users shall be able to attach images to messages or send them individually.
- The image's format must be JPEG or PNG, with a maximum file size of 6MB.
- Images get sent before messages, if they are attached to one.

8.6 Link Previews – Owner: Andrei

- For messages containing an URL, a title and brief description from the linked page are fetched.
- A card is shown under the message, containing the fetched information.
- If the link cannot be reached, no card is shown.

9. Message Interactions

9.1 Replies – Owner: Cristi

- Users can reply to any message by selecting the reply option on it.
- The reply appears as a new message with a small quoted preview of the original shown just above it.
- The quoted preview includes the original sender's name and a shortened version of the content.
- Selecting the quoted preview scrolls the conversation back to the original message.

9.2 Reactions – Owner: Cristi

- Users can react to any message with any Unicode Emojis.
- Choosing to react to a message gives the user a list of all the Emojis which they can choose from.
- Reactions are shown as Emoji counters below the message.
- Each user can only have one reaction per message and picking a different Emoji replaces the previous one.
- Selecting an existing reaction counter either adds or removes the user's own reaction of that type.

9.3 Mention System – Owner: Cristi

- Users can mention another participant in a group by typing “@” followed by their username.
- The mention appears highlighted in the sent message.

10. Moderation & Safety

10.1 Banning a Member – Owner: Eduard

- An admin can permanently ban a member from the group.
- A banned member cannot send messages or see any new messages posted after the ban.
- An admin can lift the ban at any time, restoring the member's access.

10.2 Timing Out a Member – Owner: Eduard

- An admin can give a member a temporary timeout, preventing them from sending messages until it expires.
- Available durations: 10 minutes, 1 hour, 24 hours, and 7 days, or a custom time.
- Choosing the custom time option lets the admin select a date from a calendar and time picker.
- Admins can end a timeout early at any time.

11. Read Receipts

11.1 Read Receipts in DMs – Owner: Fineas

- In a direct message, the sender can see whether the other person has read their message.
- A "Seen" indicator appears below the last message the other user has opened.

11.2 Read Receipts in Groups – Owner: Fineas

- In a group conversation, each message shows a count of how many members have read it (e.g. "Read by 4").
- When the user selects the number of people who've read their message, a list containing those users should appear.

12. Search

12.1 User Search – Owner: Rares

- When adding a member to a group, admins can search for users by username.
- People who are already in the group are excluded from the search results.

12.2 Message Search – Owner: Rares

- Within any conversation, there is a search button that opens a message search panel.
- The user can search for words or phrases within the conversation history.
- Matching messages are listed with context snippets. Selecting a result scrolls to that message.

13. Cross-Team Integrations

13.1 Community – Owner: Alex

- Users can send friend requests to anyone from the community section of the app.

13.2 Events – Owner: Alex

- Groups have the option for admins to create events.
- After an event is created, a notice appears in the chat.
- Events are exclusive to the groups that they were created in.

13.3 Matchmaking – Owner: Alex

- When the matchmaking system pairs two users together, a direct message conversation between them is created automatically if one doesn't already exist.
- A message appears in that conversation letting both users know they have matched.